

CRediT Author Statement

Counterfactual Evaluation of Causal Faithfulness in GINE Explanations for Molecular Toxicity Prediction

Gaurav Patil: Conceptualization, Methodology, Software (GNN architecture, counterfactual evaluation harness, training pipeline), Validation, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Project administration.

Parth Parmar: Methodology, Software (baseline integrations, ablation experiments, statistical verification), Validation, Formal analysis, Investigation, Writing – Review & Editing.

Both authors have read and approved the final version of the manuscript for submission to *Machine Learning with Applications*.